

Fractional Domatic, Idomatic and Total Domatic Numbers of a Graph

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Abstract

The fractional domatic number of a graph G is the maximum ratio $|\mathcal{F}|/m(\mathcal{F})$ over all families $\mathcal{F}$ of dominating sets of G , where $m(\mathcal{F})$ denotes the maximum number of times an element appears in $\mathcal{F}$. The fractional idomatic and fractional total domatic numbers are defined analogously with $\mathcal{F}$ all families of independent dominating sets and total dominating sets of G , respectively. In this chapter, we survey some results on the three parameters, and their relationship with and extension to hypergraphs.

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1 Introduction

In this chapter, we survey some results concerning the fractional domatic, fractional idomatic, and fractional total domatic numbers of a graph. First we recall the fundamental concepts of a dominating set, an independent dominating set, and a total dominating set.

A *dominating set* of a graph G is a set S of vertices of G such that every vertex not in S has a neighbor in S . The *domination number* of G , denoted $\gamma(G)$, is the minimum cardinality of a dominating set.

An *independent dominating set* of G is a set that is both a dominating set and an independent set. The *independent domination number*, denoted $i(G)$, of G is the minimum cardinality of an independent dominating set in G . An independent set of vertices in a graph G is a dominating set of G if and only if it is a maximal independent set. Thus, $i(G)$ is equivalently the minimum cardinality of a maximal independent set of vertices in G . A survey on independent domination in graphs can be found in [9].

A *total dominating set*, abbreviated TD-set, of a graph G with no isolated vertex is a set S of vertices such that every vertex in G is adjacent to a vertex in S . The *total domination number*, denoted by $\gamma_t(G)$, of G is the minimum cardinality of a TD-set of G . For a recent book on total domination in graphs, we refer the reader to [14].

The parameters studied in this chapter represent the fractional relaxation of the count of the maximum number of disjoint dominating, independent dominating, and total dominating sets. We discuss these in the next three sections. In Section 5 we discuss a common framework in hypergraphs and in Section 6 some generalizations.

2 The Fractional Domatic Number

In this section, we survey results on the fractional domatic number of a graph. First we recall the concept of the domatic number of a graph. The maximum number of vertex-disjoint dominating sets in a graph G is called the *domatic*

number of G denoted by $\text{dom}(G)$. The domatic number was introduced in 1975 by Cockayne and Hedetniemi [5] and has since been the subject of a large number of publications; a rough estimate says that it occurs in more than 200 papers to date.

Zelinka [32] wrote an excellent survey on the domatic number of a graph and its variants. As remarked by Zelinka, the word “domatic” was created from the words “dominating” and “chromatic” since, although it is defined using the concept of domination in graphs, it is somewhat analogous to the chromatic number of a graph, where we partition the vertex set into classes having certain properties (in this case, each class is a dominating set).

We consider here the fractional analogue of the parameter $\text{dom}(G)$. The *fractional domatic number* of a graph G , denoted $\text{FDOM}(G)$, is defined as

$$\text{FDOM}(G) = \max \frac{|\mathcal{F}|}{m(\mathcal{F})},$$

where the maximum is taken over all families $\mathcal{F}$ of dominating sets of G . (For a discussion of why maximum can be used instead of supremum, see Section 5.) The fractional domatic number seems to have been formally introduced in 2006 by Suomela [27], although the concept was studied in 2000 by Fujita, Yamashita, and Kameda [8].

We start with the following immediate lower and upper bounds on the fractional domatic number of a graph.

Theorem 1 *For every graph G we have*

$$\text{dom}(G) \leq \text{FDOM}(G) \leq \frac{n(G)}{\gamma(G)}.$$

Proof. To prove the lower bound, consider a family $\mathcal{F}$ that consists of a maximum number of vertex-disjoint dominating sets of G . In this case, $|\mathcal{F}| = \text{dom}(G)$ and $m(\mathcal{F}) = 1$, implying that $\text{FDOM}(G) \geq |\mathcal{F}|/m(\mathcal{F}) = \text{dom}(G)$.

To prove the upper bound, let $\mathcal{F}$ be a family of dominating sets of G . Each set in the family $\mathcal{F}$ has size at least $\gamma(G)$. Thus

$$\gamma(G) \cdot |\mathcal{F}| \leq \sum_{F \in \mathcal{F}} |F| \leq n(G) \cdot m(\mathcal{F}),$$

or, equivalently, $|\mathcal{F}|/m(\mathcal{F}) \leq n(G)/\gamma(G)$, whence the result. $\square$

We note that equality occurs in the upper bound of Theorem 1 for all complete graphs. We show next that equality also occurs in the upper bound for all cycles. We use the standard notation $[k] = \{1, \dots, k\}$.

Observation 2 *For $n \geq 3$, we have $\text{FDOM}(C_n) = n/\gamma(C_n)$.*

Proof. Let G be the cycle $v_1v_2 \dots v_nv_1$. Let S be an arbitrary minimum dominating set of G ; so $|S| = \gamma(G)$. Let $\mathcal{S} = \{j \mid j \in [n] \text{ and } v_j \in S\}$. For $i \in [n]$, let

$$S_i = \{v_{i+j} \mid j \in \mathcal{S}\},$$

where addition is taken modulo n . Each set S_i is a minimum dominating set of G . Let $\mathcal{F} = \{S_1, S_2, \dots, S_n\}$. We note that each vertex of G appears in exactly $\gamma(G)$ of these sets, implying that $m(\mathcal{F}) = \gamma(G)$ and therefore that

$$\frac{|\mathcal{F}|}{m(\mathcal{F})} = \frac{n}{\gamma(G)}.$$

Hence, $\text{FDOM}(G) \geq n/\gamma(G)$. The desired result now follows from the upper bound of Theorem 1. $\square$

The above observation immediately generalizes to any circulant. Recall that $\gamma(C_n) = \lceil n/3 \rceil$. Hence as a consequence of Observation 2, the fractional domatic number of a cycle is determined as follows.

$$\text{FDOM}(C_n) = \begin{cases} 3 & \text{if } n \equiv 0 \pmod{3} \\ \frac{3n}{n+2} & \text{if } n \equiv 1 \pmod{3} \\ \frac{3n}{n+1} & \text{if } n \equiv 2 \pmod{3} \end{cases}$$

Theorem 3 *If a graph G of order n has minimum degree δ , then*

$$\frac{n}{n-\delta} \leq \text{FDOM}(G) \leq \delta + 1.$$

Proof. Let v be a vertex of minimum degree in G , and so $d_G(v) = \delta$. Let $\mathcal{F}$ be a family of dominating sets of G . Each set in the family $\mathcal{F}$ must contain the vertex v or a neighbor of v in order to dominate v . By the Pigeonhole Principle,

at least one vertex in the closed neighborhood of v appears in at least $|\mathcal{F}|/(\delta+1)$ sets, and so $m(\mathcal{F}) \geq |\mathcal{F}|/(\delta+1)$, or, equivalently, $|\mathcal{F}|/m(\mathcal{F}) \leq \delta+1$. This is true for every family $\mathcal{F}$ of dominating sets of G , implying that $\text{FDOM}(G) \leq \delta+1$.

To prove the lower bound, consider the collection $\mathcal{F}$ of all subsets F of the vertex set of exactly $n - \delta$ elements. Since every vertex belongs to F or has a neighbor in F , each such subset F is a dominating set of H ; that is, $\mathcal{F}$ is a family of dominating sets of G . Further, every vertex is in $\binom{n-1}{n-\delta-1}$ sets of $\mathcal{F}$, and so

$$\frac{|\mathcal{F}|}{m(\mathcal{F})} = \frac{\binom{n}{n-\delta}}{\binom{n-1}{n-\delta-1}} = \frac{n}{n-\delta}.$$

Hence, $\text{FDOM}(G) \geq n/(n-\delta)$. $\square$

Thus, for example, by Theorem 3, if a graph G contains an isolated vertex, then $\text{FDOM}(G) = 1$. However, if G has no isolated vertex and S is a maximal independent set in G , then both S and $V(G) \setminus S$ are dominating sets, implying that $\text{dom}(G) \geq 2$. Thus, $\text{FDOM}(G) \geq 2$ if and only if G contains no isolated vertex.

It remains an open problem to characterize the graphs G achieving equality in the upper bound of Theorem 3. In the special case when the graph is a regular graph, Fujita, Yamashita, and Kameda proved in [8] the following result.

Theorem 4 ([8]) *If G is a δ -regular graph of order n , then $\text{FDOM}(G) = \delta + 1$ if and only if $\text{dom}(G) = \delta + 1$.*

We note that the parameter FDOM is monotonic, in that it cannot decrease on the addition of edges. We next examine the fractional domatic number of the disjoint union or join of graphs. Let G and H be two graphs. The *join* of G and H , written $G \oplus H$, is the graph obtained from the disjoint union of G and H by joining each vertex of G to every vertex of H , while the *union* of G and H , written $G + H$, is the graph consisting of the disjoint union of G and H . As a consequence of Lemma 32 later, we get the following result.

Theorem 5 *For graphs G and H it holds that $\text{FDOM}(G+H) = \min(\text{FDOM}(G), \text{FDOM}(H))$.*

A *universal vertex* of a graph G is one adjacent to all other vertices in G .

Lemma 6 *Let G and H be graphs without a universal vertex with orders n_1 and n_2 respectively.*

- (a) $\text{FDOM}(G \oplus H) = n_1$ if $n_1 = n_2$; and
- (b) $\min(n_1, n_2) < \text{FDOM}(G \oplus H) \leq (n_1 + n_2)/2$ otherwise.

Proof. The upper bound follows from Theorem 1, since the join of G and H has domination number 2. For the lower bound in (a), let $\mathcal{F}$ consist of n_1 disjoint pairs, each pair containing one vertex of G and one vertex of H . For the lower bound in (b), assume $n_1 < n_2$. Then let $\mathcal{F}$ consist of (i) all pairs of one vertex in G and one vertex in H , and (ii) the set $V(H)$. Then $|\mathcal{F}| = n_1 n_2 + 1$ and $m(\mathcal{F}) = n_2$, whence the result. $\square$

Fujita, Yamashita, and Kameda proved the following surprising and beautiful result in [8].

Theorem 7 ([8]) *If G is a cubic graph, then G has a family $\mathcal{F}$ of five dominating sets of G such that $m(\mathcal{F}) \leq 2$. Further, such a family $\mathcal{F}$ can be constructed in polynomial time.*

As an immediate consequence of Theorem 7, we have the following lower bound on the fractional domatic number of a cubic graph.

Theorem 8 *If G is a cubic graph, then $\text{FDOM}(G) \geq \frac{5}{2}$.*

As remarked by Fujita et al. [8], it is not true that if G is a cubic graph, then G has a family $\mathcal{F}$ of six dominating sets of G such that every vertex is in at most two of these. We note that if a cubic graph had such a property, then this would imply that $\text{FDOM}(G) \geq 6/2 = 3$. However, that in turn would imply that $\gamma(G) \leq n(G)/3$, but Kostochka and Stodolsky [20] showed there are cubic graphs G where $\gamma(G) = 8n(G)/23 + o(1)$.

Abbas, Egerstedt, Liu, Thomas, and Whalen [1] generalized Theorem 7 to a larger class of graphs, namely the class of $K_{1,6}$ -free graphs; that is, graphs with no induced subgraph isomorphic to $K_{1,6}$. Their study was motivated by a problem encountered both in the multiagent robotics and mobile sensor networks

domains. As remarked in [1], the generalization to $K_{1,6}$ -free graphs is of interest in multiagent robotics, because the class of $K_{1,6}$ -free graphs includes the class of unit disk graphs, where for every vertex there is a disk representing its transmission or interaction range centered at the vertex (see [24]).

In order to state their result, we recall a 1989 result due to McCuaig and Shepherd [23]. Let $\mathcal{B}$ be the family of seven graphs shown in Figure 1. McCuaig and Shepherd [23] showed that if G is a connected graph of minimum degree at least 2 and G is not one of graphs in the family $\mathcal{B}$, then the domination number of G is at most two-fifths its order.

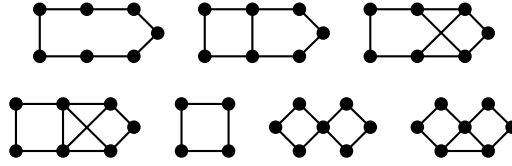


Figure 1. The family $\mathcal{B}$ of seven exceptional graphs

We are now in a position to state the result of Abbas et al. [1]:

Theorem 9 ([1]) *If G is a $K_{1,6}$ -free connected graph with minimum degree at least 2 and $G \notin \mathcal{B} \cup \{K_{2,3}\}$, then G has a family $\mathcal{F}$ of five dominating sets of G such that $m(\mathcal{F}) \leq 2$, implying that $\text{FDM}(G) \geq \frac{5}{2}$.*

We remark that the proof of Theorem 9 given by Abbas et al. [1] is algorithmic and gives a polynomial-time algorithm to find such a family $\mathcal{F}$.

We conclude this section with a comment about planar graphs. By Theorem 3, the maximum fractional domatic number of a planar graph is at most 6, since the minimum degree is at most 5. As observed in [12], an example of a planar graph with fractional domatic number equal to 6 is the icosahedron; this has six disjoint independent dominating sets of size 2, each consisting of a vertex and the unique vertex at distance 3 from it.

3 The Fractional Idomatic Number

In this section, we survey results on the fractional idomatic number of a graph. First we recall the concept of the idomatic number of a graph. The maximum number of vertex disjoint independent dominating sets in a graph G is called the *idomatic number* of G denoted by $\text{idom}(G)$. This terminology was introduced by Zelinka [30], but the parameter was originally defined by Cockayne and Hedetniemi [6]. In this section we consider a fractional version of the idomatic number of a graph. The *fractional idomatic number* of graph G , denoted $\text{FIDOM}(G)$, is defined as

$$\text{FIDOM}(G) = \max \frac{|\mathcal{F}|}{m(\mathcal{F})}$$

where the maximum is taken over all families $\mathcal{F}$ of independent dominating sets of G and where, as before, $m(\mathcal{F})$ is the maximum number of times an element appears in $\mathcal{F}$. (We note that this parameter should be defined as the supremum, but as explained in Section 5 one can show that the supremum is achieved.) The results in this section are mainly due to the authors [12].

Using analogous proofs to those presented in Theorem 1 and 3, one can establish the following immediate lower and upper bounds on the fractional idomatic number of a graph.

Theorem 10 ([12]) *The following hold in a graph G with minimum degree δ .*

- (a) $\text{idom}(G) \leq \text{FIDOM}(G) \leq n(G)/i(G)$.
- (b) $\text{FIDOM}(G) \leq \delta + 1$.

Examples of equality in the upper bound of Theorem 10(a) are the cycles of length a multiple of 3. We note that $\text{FIDOM}(G) = 1$ only if the graph G has an isolate. As shown, for example, by Favaron [7], the independent domination number of a graph of order n can be as much as $n - o(n)$, even with prescribed minimum degree. For such a graph G , we have that $\text{idom}(G) = 1$ and that $\text{FIDOM}(G) = 1 + o(1)$. Even restricted to special classes of graphs, one can still see this behavior. Here is one such result. Recall that a graph is *claw-free* if it does not contain $K_{1,3}$ as an induced subgraph.

Observation 11 ([12]) *There exist connected claw-free graphs G with arbitrarily large minimum degree for which $\text{FIDOM}(G) = 1 + o(1)$.*

Proof. Let G be the claw-free graph G constructed as follows. For a and d positive integers, let G be obtained from a complete graph H of order ad as follows. Let $X_1, \dots, X_a$ be a partition of $V(H)$ into sets each of size d . For each resulting set X_i , we add a vertex x_i of degree d adjacent to every vertex of X_i for $i \in [a]$. The resultant split graph G is claw-free and has minimum degree d . Further letting $X = \{x_1, \dots, x_a\}$, we note that every independent dominating set of G contains at least $a - 1$ vertices of X , implying that

$$\text{FIDOM}(G) \leq \frac{a}{a-1}.$$

Further, for all $i \in [a]$ if $F_i = (X \setminus \{x_i\}) \cup \{x'_i\}$ where x'_i is an arbitrary vertex in X_i , then $\mathcal{F} = \{F_1, \dots, F_a\}$ is a family of independent dominating sets of G satisfying $m(\mathcal{F}) = a - 1$, implying that

$$\text{FIDOM}(G) \geq \frac{|\mathcal{F}|}{m(\mathcal{F})} = \frac{a}{a-1}.$$

Consequently, $\text{FIDOM}(G) = a/(a-1)$. The desired result now follows by taking a and d arbitrarily large. $\square$

We next present a lower bound on the fractional idiomatic number in terms of dynamic colorings. An r -dynamic coloring, also called r -hued coloring in the literature, of a graph G is a proper coloring of the vertices of G such that every vertex v has at least $\min(d_G(v), r)$ colors in its neighborhood, where $d_G(v)$ is the degree of the vertex v in G . For more details on r -dynamic colorings, we refer the reader to Jahanbekam et al. [18].

Theorem 12 ([12]) *If G is a graph with minimum degree at least r that has an r -dynamic coloring using k colors, then*

$$\text{FIDOM}(G) \geq \frac{k}{k-r}$$

and therefore $i(G) \leq (k-r)n/k$.

As a consequence of Theorem 12, we have the following lower bound on the fractional idiomatic number in terms of the chromatic number.

Theorem 13 ([12]) *If G is an isolate-free graph with chromatic number k , then*

$$\text{FIDOM}(G) \geq \frac{k}{k-1}.$$

By Theorem 10(b), the maximum fractional idomatic number of a planar graph is at most 6, since the minimum degree is at most 5. As noted earlier, an example of a planar graph with fractional idomatic number equal to 6 is the icosahedron; this has six disjoint independent dominating sets of size 2, each consisting of a vertex and the unique vertex at distance 3 from it.

We next consider lower bounds on the fractional idomatic number of a planar graph. As a consequence of Theorem 13 and a result by MacGillivray and Seyfarth [21] that provides an upper bound for the independent domination number in terms of the chromatic number, we obtain the following result.

Theorem 14 ([12]) *The following hold in a planar connected graph G of order n .*

- (a) *If $n \geq 2$, then $\text{FIDOM}(G) \geq \frac{4}{3}$.*
- (b) *If $n \geq 10$, then $i(G) \leq \frac{3}{4}n - 2$.*

Recall that the *corona* $G \circ P_1$ of a graph G , also denoted $\text{cor}(G)$ in the literature, is the graph obtained from G by adding a pendant edge to each vertex of G . As remarked in [12], the two bounds in Theorem 14 are sharp because of the corona $K_4 \circ P_1$ of K_4 illustrated in Figure 2.

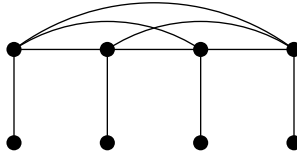


Figure 2. A planar graph G with $\text{FIDOM}(G) = \frac{4}{3}$

If one considers planar graphs of minimum degree at least 2, then the lower bound on the fractional idomatic number in Theorem 14(a) can be improved. The key is a result due to Kim, Lee and Park [19], who showed that every connected planar graph has a 2-dynamic coloring using at most four colors, except for C_5 . Therefore by Theorem 12 (with $r = 2$ and $k = 4$), we have that every

connected planar graph G , except possibly for C_5 , satisfies $\text{FIDOM}(G) \geq 2$. However, since the 5-cycle has $\text{FIDOM}(C_5) = \frac{5}{2}$, the 5-cycle is no exception to the lower bound $\text{FIDOM}(G) \geq 2$. We therefore have the following lower bound on the fractional idiomatic number of a planar graph with minimum degree at least 2.

Theorem 15 ([12]) *If G is a planar graph with $\delta(G) \geq 2$, then $\text{FIDOM}(G) \geq 2$.*

As an immediate consequence of Theorem 10(a) and Theorem 15, we have the following upper bound on the independent domination number of a planar graph with minimum degree at least 2.

Corollary 16 ([12]) *If G is a planar graph with $\delta(G) \geq 2$, then $i(G) \leq \frac{1}{2}n(G)$.*

The following construction, given in [12], shows that there exists an infinite family of planar graphs G with minimum degree two that satisfy $\text{FIDOM}(G) = 2$. For $s \geq 2$, let H_s be the graph obtained from a 4-cycle $v_1v_2v_3v_4v_1$ by adding s new vertices whose neighbors are the pair $\{v_i, v_{i+2}\}$ for each $i \in [2]$. When $s = 4$, for example, the resulting graph H_s is illustrated in Figure 3. As observed in [12], it holds that $i(H_s) = \frac{1}{2}n(H_s)$ and $\text{FIDOM}(H_s) = 2$. Thus, the bounds of Theorem 15 and Corollary 16 are tight.

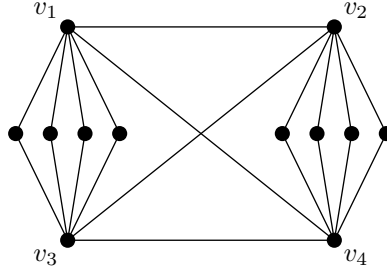


Figure 3. The planar graph H_4 with $i(H_4) = \frac{1}{2}n(H_4)$ and $\text{FIDOM}(H_4) = 2$

As remarked in [12], there are numerous families of planar graphs G with minimum degree two that satisfy $\text{FIDOM}(G) = 2$ and $i(G) < \frac{1}{2}n(G)$. It remains, however, an open problem to characterize the graphs achieving equality in the bounds of Theorem 15 and Corollary 16. It is also noted in [12] that there are

numerous families of planar graphs G with minimum degree two that do not have two disjoint independent dominating sets, and therefore such graphs G satisfy $\text{idom}(G) = 1$ and $\text{FIDOM}(G) \geq 2$.

It remains an open problem to determine a best possible lower bound on the fractional idiomatic number of a planar graph with minimum degree 3. In this case, we believe that the upper bound of Theorem 15 can be improved from 2 to $\frac{5}{2}$. We pose this formally as a conjecture.

Conjecture 1 *If G is a planar graph with $\delta(G) \geq 3$, then $\text{FIDOM}(G) \geq \frac{5}{2}$.*

We note that if $G = C_5 \square K_2$ is the 5-prism illustrated in Figure 4, then $i(G) = 4 = \frac{2}{5}n$ and $\text{FIDOM}(G) = \frac{5}{2}$. Thus if Conjecture 1 is true, the bound is best possible.

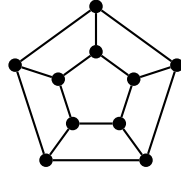


Figure 4. The 5-prism $C_5 \square K_2$.

The fractional idiomatic number of a maximal outerplanar graph is easy to compute. If G is a maximal outerplanar graph, then G is 3-colorable and every color class is an independent dominating set, implying that $\text{idom}(G) \geq 3$. Further since G has minimum degree 2, Theorem 10(b) implies that $\text{FIDOM}(G) \leq 3$. Consequently, $\text{idom}(G) = \text{FIDOM}(G) = 3$ for a maximal outerplanar graph G , as observed in [12].

If G is a general outerplanar graph, then G is 3-chromatic and, by Theorem 13, we therefore have $\text{FIDOM}(G) \geq \frac{3}{2}$. This in turn implies by Theorem 10(a) that $i(G) \leq \frac{2}{3}n(G)$. If G is obtained from K_3 by attaching $k \geq 1$ pendant edges to each vertex of the triangle, then the resulting graph G has order $n(G) = 3(k+1)$ and satisfies $i(G) = 2k+1$. Thus for k sufficiently large, the bounds $i(G) \leq \frac{2}{3}n(G)$ and $\text{FIDOM}(G) \geq \frac{3}{2}$ are asymptotically sharp. If, however, we impose a minimum degree condition, then we can improve the lower bound of $\frac{3}{2}$ on the fractional idiomatic number to a lower bound of 2, as shown by the following result in [12].

Theorem 17 ([12]) *If G is an outerplanar graph with minimum degree at least 2, then $\text{FIDOM}(G) \geq \text{idom}(G) \geq 2$.*

Let G and H be two graphs. As above the *join* of G and H is written $G \oplus H$ while their disjoint *union* is written $G + H$. The *lexicographic product* of G and H , written $G[H]$, is the graph with vertex set $V(G) \times V(H)$ where two vertices (g_1, h_1) and (g_2, h_2) are adjacent in $G[H]$ if and only if $g_1 g_2 \in E(G)$ or $g_1 = g_2$ and $h_1 h_2 \in E(H)$. For these graph operations, the fractional idiomatic number behaves as follows.

Theorem 18 ([12]) *For graphs G and H , the following hold.*

- (a) $\text{FIDOM}(G + H) = \min(\text{FIDOM}(G), \text{FIDOM}(H))$.
- (b) $\text{FIDOM}(G \oplus H) = \text{FIDOM}(G) + \text{FIDOM}(H)$.
- (c) $\text{FIDOM}(G[H]) = \text{FIDOM}(G) \times \text{FIDOM}(H)$.

The following result is shown in [11].

Theorem 19 ([11]) *If G is a graph with minimum degree at least 2 and maximum degree at most 3, then $\text{idom}(G) \geq 2$.*

As an immediate consequence of Theorem 19, every cubic graph G satisfies $\text{idom}(G) \geq 2$, a result attributed to Berge. We note that there are many cubic graphs G satisfying $\text{idom}(G) = 2$. We pose the following question.

Question 1 *Is it true that if G is a connected cubic graph different from $K_{3,3}$, then $\text{FIDOM}(G) \geq \frac{5}{2}$?*

As observed earlier, if $G = C_5 \square K_2$ is the 5-prism shown in Figure 4, then $\text{FIDOM}(G) = \frac{5}{2}$. Hence if Question 1 is true, then the lower bound value $5/2$ for the fractional idiomatic number would be best possible.

4 The Fractional Total Domatic Number

In this section, we survey results on the fractional total domatic number of a graph. First we recall the concept of the total domatic number of a graph. The *total domatic number* of a graph G , denoted by $tdom(G)$ and first defined by Cockayne, Dawes, and Hedetniemi [4], is the maximum number of total dominating sets into which the vertex set of G can be partitioned. The parameter $tdom(G)$ is equivalent to the maximum number of colors in a (not necessarily proper) coloring of the vertices of a graph where every color appears in every open neighborhood. Chen, Kim, Tait, and Verstraete [3] called this the *coupon coloring problem*. This parameter is now well studied. We refer the reader to Chapter 13 in the book [14] on total domination in graphs for a survey of results on the total domatic number, and to [10] for a recent paper on this topic.

In this section we consider a fractional version of the total domatic number of a graph. The *fractional total domatic number* of a graph G , denoted $FTD(G)$, is defined as

$$FTD(G) = \max \frac{|\mathcal{F}|}{m(\mathcal{F})}$$

where the maximum is taken over all families $\mathcal{F}$ of total dominating sets of G and where, as before, $m(\mathcal{F})$ is the maximum number of times an element appears in $\mathcal{F}$. (As before the parameter should be defined as the supremum, but as shown in Section 5, the supremum is achieved.)

The following trivial lower and upper bounds on the fractional total domatic number of a graph are established in [10].

Theorem 20 ([10]) *If G is an isolate-free graph, then*

$$tdom(G) \leq FTD(G) \leq \frac{n(G)}{\gamma_t(G)}.$$

Theorem 21 ([10]) *If a graph G of order n has minimum degree $\delta \geq 1$, then*

$$\frac{n}{n - \delta + 1} \leq FTD(G) \leq \delta.$$

Thus, for example, by Theorems 20 and 21 we get the following observation.

Observation 22 ([10]) *The following hold in a graph G with minimum degree δ .*

- (a) *If $\delta \geq 1$, then $FTD(G) = 1$.*
- (b) *If $\delta \geq 2$, then $FTD(G) > 1$.*

We note that there are graphs G with arbitrarily large minimum degree with $FTD(G) < 1 + \epsilon$ for any given $\epsilon > 0$. Indeed, these are the graphs that Zelinka [31] provided as examples that have $tdom(G) = 1$ and arbitrarily large minimum degree.

A lower bound on the fractional total domatic number of a claw-free graph with minimum degree at least 2 is determined in [10].

Theorem 23 ([10]) *If G is a claw-free graph with $\delta \geq 2$, then $FTD(G) \geq \frac{3}{2}$.*

The lower bound of Theorem 23 is in a sense best possible in that the graphs K_3 and C_6 have fractional total domatic number exactly $3/2$. However, asymptotically the bound should be improvable. As remarked in [10], perhaps it is true that if G is a connected, claw-free graph with $\delta \geq 2$, then $FTD(G) \geq 2 - o(1)$. Further, if G is a connected, claw-free graph with $\delta \geq 3$, then maybe this guarantees that $FTD(G) \geq 2$. As shown in [10], there are arbitrarily large connected $K_{1,4}$ -free graphs with fractional total domatic number exactly $3/2$.

We note that the union of two disjoint dominating sets is a total dominating set. Thus, it is immediate that $tdom(G) \geq \lfloor dom(G)/2 \rfloor$. But, in the case that the ordinary domatic number is odd, one can say slightly more:

Theorem 24 ([10]) *If G is an isolate-free graph, then $FTD(G) \geq \frac{1}{2} dom(G)$.*

Proof. Let $dom(G) = k$, and let $D_1, \dots, D_k$ be k disjoint dominating sets in the graph G . The family

$$\mathcal{F} = \{ D_i \cup D_j \mid 1 \leq i < j \leq k \}$$

is a family of $\binom{k}{2}$ total dominating sets of G . Since every vertex of G appears in at most $k - 1$ sets in the family $\mathcal{F}$, we note that $m(\mathcal{F}) = k - 1$. Thus,

$$FTD(G) \geq \frac{|\mathcal{F}|}{m(\mathcal{F})} = \frac{\binom{k}{2}}{k - 1} = \frac{k}{2}.$$

The desired result now follows recalling that $k = \text{dom}(G)$. $\square$

Equality in Theorem 24 occurs, for example, in complete graphs.

We consider now planar graphs. A *triangulated disc* is a (simple) planar graph all of whose faces are triangles, except possibly for the outer face. Matheson and Tarjan [22] showed that if G is a triangulated disc, then $\text{dom}(G) \geq 3$. Hence, as an immediate consequence of Theorem 24 and the Matheson-Tarjan result, we have the following lower bound on the fractional total domatic number of a triangulated disc.

Theorem 25 ([10]) *If G is a triangulated disc, then $\text{FTD}(G) \geq \frac{3}{2}$.*

As remarked in [10], the lower bound of Theorem 25 is tight as may be seen by considering the triangulated disc G illustrated in Figure 5, where the shaded area consists of any maximal planar graph (or, equivalently, triangulation). Let S be the set of three vertices on the outer face of G that have degree at least 4. If $\mathcal{F}$ is an arbitrary family of total dominating sets of G , then each set in the family $\mathcal{F}$ contains at least two vertices of S . By averaging there is a vertex in S that belongs to at least $2|\mathcal{F}|/3$ sets in $\mathcal{F}$, implying that $m(\mathcal{F}) \geq 2|\mathcal{F}|/3$, or, equivalently, $|\mathcal{F}|/m(\mathcal{F}) \leq 3/2$. Since this is true for every family $\mathcal{F}$ of total dominating sets of G , this implies that $\text{FTD}(G) \leq 3/2$. Consequently, by Theorem 25, $\text{FTD}(G) = 3/2$.

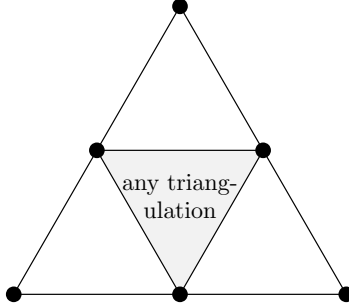


Figure 5. A triangulated disc G with $\text{FTD}(G) = \frac{3}{2}$

We next consider triangulations, where by *triangulation* we mean a simple graph embedded in some orientable surface such that every region is a triangle.

The following key lemma establishes an upper bound on the fractional total domatic number of a graph G in terms of its average degree, which we denote by $\bar{d}_{av}(G)$.

Lemma 26 ([10]) *If G is a triangulation of order at least 4, then*

$$FTD(G) \leq \bar{d}_{av}(G) - 1.$$

If G is a planar triangulation of order n , then $\bar{d}_{av}(G) = 6 - \frac{12}{n}$. Thus, as an immediate consequence of Lemma 26, we have the following upper bounds on the total domatic and fractional total domatic numbers of a planar graph.

Theorem 27 ([10]) *The following hold in a planar graph G .*

- (a) $tdom(G) \leq 4$.
- (b) $FTD(G) \leq 5 - \frac{12}{n}$.

As remarked in [10] there are planar graphs G with $tdom(G) = 4$. For example, if G is obtained from a truncated tetrahedron and adding a vertex inside each hexagonal face that is joined to all vertices on the boundary, then G is a planar graph G of order 16 satisfying $tdom(G) = 4$. Illustrated in Figure 6 (which corrects a figure in [10]) is a spanning subgraph thereof that still has four disjoint total dominating sets: the vertices labelled i form a total dominating set for each $i \in [4]$.

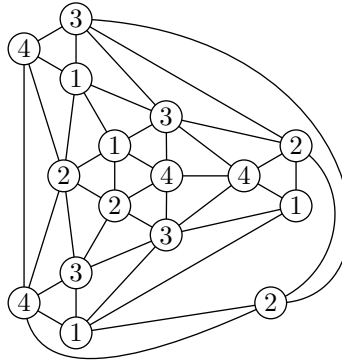


Figure 6. A planar graph G with $tdom(G) = 4$

As shown in [10] there are planar graphs G for which $FTD(G) > 4$. We note that the result of Lemma 26 applies on all surfaces. In particular, since the average degree of a toroidal graph is at most 6, this yields the following upper bounds on the total domatic and fractional total domatic numbers of a toroidal graph.

Theorem 28 *If G is a toroidal graph, then $tdom(G) \leq FTD(G) \leq 5$.*

There are toroidal graphs G satisfying $tdom(G) = FTD(G) = 5$. The example provided in [10] is illustrated in Figure 7, where the top and bottom dotted lines should be identified and similarly with the left and right dotted lines; the vertices labelled i form a total dominating set of G for each $i \in [5]$.

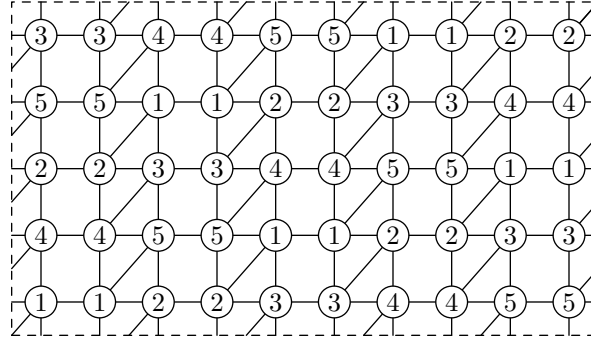


Figure 7. A toroidal graph G with $tdom(G) = 5$

It remains generally an open problem to determined good lower bounds for triangulations. Since every planar triangulation is a triangulated disc, Theorem 25 implies that every planar triangulation G satisfies $FTD(G) \geq \frac{3}{2}$. However we believe this lower bound can be improved. In this regard, the following conjectures are posed in [10].

Conjecture 2 ([10]) *If G is a planar triangulation of order at least 4, then $tdom(G) \geq 2$.*

Using the Four Color Theorem, it is shown in [10] that Conjecture 2 is true if every vertex has odd degree or if the dual of G is Hamiltonian. By characterizing the maximal outerplanar graphs H that have $tdom(H) < 2$, Nagy [25] showed

that the conjecture is true if G is Hamiltonian. A stronger version of the above is the following conjecture.

Conjecture 3 ([10]) *Every planar triangulation with at least four vertices has a proper 4-coloring (C_1, C_2, C_3, C_4) such that $C_1 \cup C_2$ and $C_3 \cup C_4$ are total dominating sets.*

As with the parameter FDM , the parameter FTD is monotonic. The fractional total domatic number of the disjoint union of graphs behaves as expected.

Theorem 29 ([10]) *For all graphs G and H it holds that $\text{FTD}(G + H) = \min\{\text{FTD}(G), \text{FTD}(H)\}$.*

We conclude this section with some comments on regular graphs. It remains a long-standing open problem to characterize those cubic graphs that have two disjoint total dominating sets; that is, the 3-regular graphs with $\text{tdom}(G) = 1$. It is well known that the Heawood graph, shown in Figure 8, is the smallest example of such a graph G without two disjoint total dominating sets.

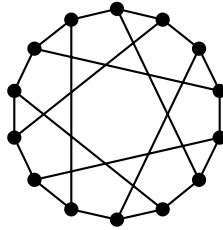


Figure 8. The Heawood graph

Nevertheless, a natural question is whether the fractional total domatic number of a cubic graph is always at least 2. This question was answered in [16].

Theorem 30 ([16]) *If G is a connected cubic graph, then $\text{FTD}(G) \geq 2$.*

Recall that Fujita, Yamashita, and Kameda proved in Theorem 7 the beautiful result that every connected cubic graph G has a family $\mathcal{F}$ of five dominating sets such that every vertex is in at most two of these. The following strengthening of Theorem 30 is conjectured in [10].

Conjecture 4 ([10]) *If G is a connected cubic graph, then G has a family of four total dominating sets of G such that every vertex is in at most two of these.*

For regular graphs with higher minimum degree, we have the following lower bound on the fractional total domatic number of a regular graph.

Theorem 31 ([16]) *For all $k \geq 3$, if G is a k -regular graph, then*

$$FTD(G) > \frac{k}{1 + \ln(k)}.$$

5 Fractional Definitions and Hypergraphs

The three parameters explored in this chapter can be defined in terms of hypergraphs. The *fractional matching number* of a hypergraph H , denoted $\nu^*(H)$, is defined by the linear program

$$\max \sum_{e \in E(H)} w(e) \quad \text{such that} \quad \forall e \in E : w(e) \geq 0 \quad \text{and} \quad \forall v \in V(H) : \sum_{e \ni v} w(e) \leq 1.$$

The *matching number* of H , denoted $\nu(H)$, is the maximum number of disjoint hyperedges. Clearly $\nu^*(H) \geq \nu(H)$. Also, by linear programming duality, the fractional matching number equals the fractional transversal/cover number.

As before, for a multisubset $\mathcal{F}$ of $E(H)$, we define $m(\mathcal{F})$ as the maximum number of times a vertex of H appears in $\mathcal{F}$. Since the linear program has a rational solution,

$$\nu^*(H) = \sup \frac{|\mathcal{F}|}{m(\mathcal{F})} = \max \frac{|\mathcal{F}|}{m(\mathcal{F})},$$

where the maximum and the supremum is over all such multisets $\mathcal{F}$. See Chapter 1 of [26] for a fuller discussion.

Consider a family $\mathfrak{D}$ of subsets of the vertex set of a graph G . This set can naturally be thought of as a hypergraph $H_{\mathfrak{D}}$. Then the matching number of $H_{\mathfrak{D}}$ is the maximum number of disjoint members of $\mathfrak{D}$. If we let $\mathfrak{D}$ be the set of dominating sets, we get the domatic number and fractional domatic number, discussed in Section 2. If we let $\mathfrak{D}$ be the set of independent dominating sets, we get the idomatic number and fractional idomatic number, discussed in Section 3.

If we let $\mathfrak{D}$ be the set of total dominating sets, we get the total domatic number and fractional total domatic number, discussed in Section 4.

The results about the behavior of the three fractional parameters under disjoint union are a special case of a result in hypergraphs. Given two disjoint hypergraphs H_1 and H_2 , we define their *direct sum* as the hypergraph with vertex set $V(H_1) \cup V(H_2)$ and edge set $\{e_1 \cup e_2 \mid e_1 \in E(H_1), e_2 \in E(H_2)\}$. Bujtás and Tuza [2] showed that the matching number of the direct sum of two hypergraphs equals the smaller of the two matching numbers. We note that the analogous result is true for the fractional matching number too.

Lemma 32 *If hypergraph H is the direct sum of hypergraphs H_1 and H_2 , then $\nu^*(H) = \min(\nu^*(H_1), \nu^*(H_2))$.*

Proof. For $\ell \in \{1, 2\}$, let h_ℓ be an optimal weighting of $E(H_\ell)$. Let $Y = \max(\nu^*(H_1), \nu^*(H_2))$. Then define the weighting h of the direct sum H by

$$h(e_1 \cup e_2) = h_1(e_1)h_2(e_2)/Y.$$

For each vertex $v \in V(H_1)$ we have

$$\sum_{e \ni v} h(e) = \sum_{e_1 \ni v} h_1(e_1) \sum_{e_2} h_2(e_2)/Y \leq \sum_{e_1 \ni v} h_1(e_1) \leq 1.$$

Similarly, the constraint is satisfied for $v \in V(H_2)$. And the total weight of h is $\nu^*(H_1)\nu^*(H_2)/Y = \min(\nu^*(H_1), \nu^*(H_2))$. It follows that $\nu^*(H) \geq \min(\nu^*(H_1), \nu^*(H_2))$.

Conversely, let h be the optimal weighting for the direct sum H . Then define the weighting h_1 on hypergraph H_1 by

$$h_1(e_1) = \sum_{e \supseteq e_1} h(e).$$

It follows readily that for each vertex $v \in V(H_1)$ we have $\sum_{e_1 \ni v} h_1(e_1) \leq 1$. That is, the weighting h_1 represents a fractional matching of H_1 . Thus $\nu^*(H_1) \geq \nu^*(H)$. Analogously, $\nu^*(H_2) \geq \nu^*(H)$. That is, $\nu^*(H) \leq \min(\nu^*(H_1), \nu^*(H_2))$.

The two inequalities combined give the desired result. $\square$

As a consequence follows the results on disjoint union for the fractional domatic, independent domatic and total domatic numbers given earlier.

6 More on Hypergraphs

While the previous section described a general conversion from a particular domatic number of a graph to the matching number of an associated hypergraph, another hypergraph provides a more general setting for the fractional domatic and fractional total domatic number.

Recall that a subset T of vertices is a *transversal* (also called *vertex cover* or *hitting set*) in a hypergraph H if T has a nonempty intersection with every edge of H . The *transversal number* $\tau(H)$ of H is the minimum size of a transversal in H . We denote by $\text{disj}_\tau(H)$ the *disjoint transversal number* of a hypergraph H , which is the maximum number of disjoint transversals in H . Analogous to the fractional total domatic number, one can define the fractional disjoint transversal number. The *fractional disjoint transversal number* of H , denoted $FDT(H)$, is defined as

$$FDT(H) = \max \frac{|\mathcal{F}|}{m(\mathcal{F})},$$

where the maximum is taken over all families $\mathcal{F}$ of transversals of H . Analogous to earlier results, we have the following bounds on the fractional disjoint transversal number.

Theorem 33 ([10]) *For every isolate-free hypergraph H of order n ,*

$$\text{disj}_\tau(H) \leq FDT(H) \leq \frac{n}{\tau(H)}.$$

For $k \geq 2$, if H is the complete k -uniform hypergraph of order n , then $\tau(H) = n - k + 1$, and so by Theorem 33, $FDT(H) \leq n/(n - k + 1)$. To prove that $FDT(H) \geq n/(n - k + 1)$, we consider the collection $\mathcal{F}$ of all $n - k + 1$ element subsets of $V(H)$. The resulting family $\mathcal{F}$ is a family of $\binom{n}{n-k+1}$ transversals of H . Each vertex is in $\binom{n-1}{n-k}$ sets of $\mathcal{F}$, and so

$$FDT(H) \geq \frac{|\mathcal{F}|}{m(\mathcal{F})} = \frac{\binom{n}{n-k+1}}{\binom{n-1}{n-k}} = \frac{n}{n - k + 1}.$$

Consequently, $FDT(H) = n/(n - k + 1)$. Thus the upper bound of Theorem 33 is achieved, for example, by the complete k -uniform hypergraph of order n .

As another example, if F_7 is the Fano plane, illustrated in Figure 9, then $\tau(F_7) = 3$, and so by Theorem 33, $FDT(F_7) \leq 7/3$. If we take $\mathcal{F}$ to be the family consisting of the seven edges of F_7 , then $\mathcal{F}$ is a family of transversals of F_7 . Each vertex belongs to exactly three sets of $\mathcal{F}$, and so $FDT(F_7) \geq |\mathcal{F}|/m(\mathcal{F}) = 7/3$. Consequently, $FDT(F_7) = 7/3$. Thus the Fano plane achieves equality in the upper bound of Theorem 33. We note however that $disj_\tau(F_7) = 1$.

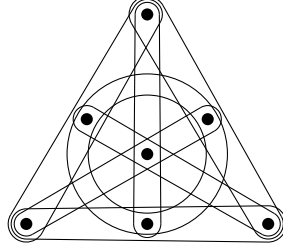


Figure 9. The Fano plane F_7

Analogously to Lemma 32, we have the following result on the fractional disjoint transversal number of the disjoint union of hypergraphs.

Theorem 34 ([10]) *If H is the disjoint union of isolate-free hypergraphs H_1 and H_2 , then $FDT(H) = \min\{FDT(H_1), FDT(H_2)\}$.*

We describe next the interplay between the fractional (total) domatic number and the fractional disjoint transversal number. The *open neighborhood hypergraph*, abbreviated ONH, of a graph G is the hypergraph $\mathcal{ON}(G)$ whose vertex set is $V(G)$ and whose hyperedges are the open neighborhoods of vertices in G . Thus, if $H = \mathcal{ON}(G)$, then $V(H) = V(G)$ and $E(H) = \{N_G(x) \mid x \in V(G)\}$. As first observed by Thomassé and Yeo [28], a total dominating set in G is a transversal in $\mathcal{ON}(G)$, and conversely. Thus, the transversal number of $\mathcal{ON}(G)$ is precisely the total domination number $\gamma_t(G)$. Similarly, the *closed neighborhood hypergraph*, abbreviated CNH, of a graph G is the hypergraph $\mathcal{CN}(G)$ whose vertex set is $V(G)$ and whose hyperedges are the closed neighborhoods of vertices in G . Again, a dominating set in G is a transversal in $\mathcal{CN}(G)$ and conversely. We state this connection formally as follows.

Observation 35 ([10]) *For every graph G , $\text{FDM}(G) = \text{FDT}(\mathcal{CN}(G))$ and for every isolate-free graph G , $\text{FTD}(G) = \text{FDT}(\mathcal{ON}(G))$.*

As an example, if G is the Heawood graph, illustrated in Figure 8, then the ONH consists of two disjoint copies of the Fano plane F_7 . So by Theorem 34 and Observation 35, we have $\text{FTD}(G) = \text{FDT}(\mathcal{ON}(G)) = \text{FDT}(F_7 \cup F_7) = \text{FDT}(F_7) = 7/3$. In contrast, the Heawood graph does not have two disjoint total dominating sets; that is, $\text{tdom}(G) = 1$. Nevertheless there is a general fractional lower bound.

Theorem 36 ([16]) *If H is a 3-regular 3-uniform hypergraph, then $\text{FDT}(H) \geq 2$, and this bound is tight.*

As regards tightness, it is shown in [13] that there are infinitely many (connected) 3-regular 3-uniform hypergraphs H satisfying $\tau(H) = \frac{1}{2}n(H)$, implying by Theorem 33 that each of these hypergraphs satisfy $\text{FDT}(H) \leq 2$. The lower bound in Theorem 36 shows that $\text{FDT}(H) \geq 2$. Consequently, $\text{FDT}(H) = 2$ for these hypergraphs.

As a consequence of a result due to Thomassen [29] and a relationship given in [15] between the total domatic number of a k -regular graph and 2-colorings of k -uniform k -regular hypergraphs, we have that if H is a 4-regular 4-uniform hypergraph, then $\text{FDT}(H) \geq \text{disj}_\tau(H) \geq 2$. However, it is conjectured in [16] that this lower can be improved when H is a 4-regular 4-uniform hypergraph.

Conjecture 5 ([16]) *If H is a 4-regular 4-uniform hypergraph, then $\text{FDT}(H) \geq \frac{7}{3}$.*

We note that if Conjecture 5 is true, then it implies that every 4-regular graph G satisfies $\text{FTD}(G) \geq \frac{7}{3}$.

Using probabilistic arguments, the following lower bound on the fractional disjoint transversal number of a k -regular k -uniform hypergraph was established in [16].

Theorem 37 ([16]) *For all $k \geq 3$, if H is a k -regular k -uniform hypergraph, then*

$$\text{FDT}(H) > \frac{k}{1 + \ln(k)}.$$

Further, this bound is essentially best possible as there exist k -regular k -uniform hypergraphs H_k with $FDT(H_k) \leq \frac{k}{\ln(k)}(1 + o(1))$.

7 Conclusion

ADD ME?

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